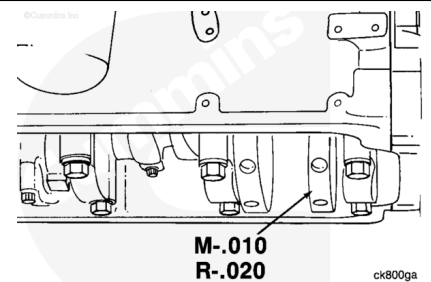


Trainings ▾

General Information

The crankshaft will be stamped on the end of the number 1 counterweight to indicate whether it has been ground undersize. The amount of material removed from the main journals and the rod journals are both stamped at this location. Thrust bearing size is stamped on a crankshaft counterweight adjacent to the thrust location.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and can cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. See equipment manufacturer service information.

- Drain the oil from the lubricating oil pan. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/89/89-007-037.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/89/89-007-025.html)
- Remove the oil suction tube and o-ring. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/89/89-001-089.html)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/89/89-001-089.html)

Remove

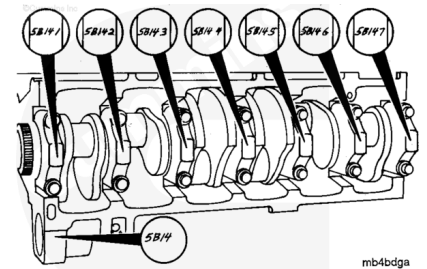


Remove the bearings one set at a time. Personal injury and damage to the crankshaft can result if the crankshaft falls.

The main bearing caps **must** be marked for position. The last number on each bearing cap identifies the cap location position in the block.

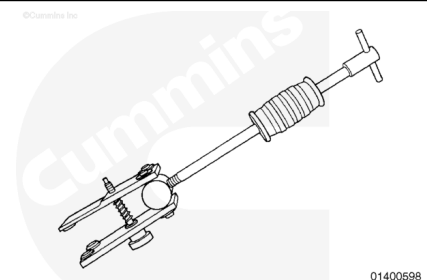
The block and main bearing cap identification numbers **must** be identical.

Use a steel stamp and mark any cap that is **not** marked correctly.



Loosen the main bearing cap capscrews until there is approximately 25 mm [1 in] between the capscrew head and the cap. Use a main bearing replacer, Part Number ST1116, to loosen the cap.

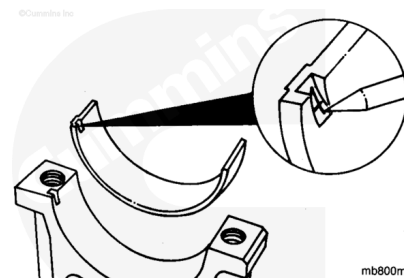
Remove the cap and capscrews.



Remove the lower bearing. Use an awl to mark the bearing position in the tang area.

Mark the bearing position for future identification or for possible failure analysis.

Use the following procedure for instructions on removal of the thrust bearing. Refer to Procedure 001-007 in Section 1.
(/qs3/pubsys2/xml/en/procedures/89/89-001-007.html)



Remove the upper main bearing shells.

Insert the main bearing roll-out tool, Part Number 3823818, into the crankshaft oil drilling.

Rotate the crankshaft so the tang (1) of the main bearing shell rolls out of the block.

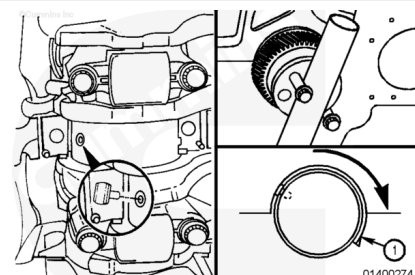
Use the crankshaft adapter mounting bolts and slowly turn the crankshaft until the bearing shell is out of the block.

Note : Crankshaft journal number 4 has no oil drilling.

For removing upper bearing number 4, use a flat piece of bar stock (approximately 3.175 x 38.1 mm [0.125 x 1.5 in]) to press the non-tang side of the bearing. It can be necessary to turn the crankshaft at the same time.

Once the bearing starts to turn out of the block, use a pry bar to gently pry between the block and the back side of the bearing.

While applying pressure to the back side of the bearing, turn the crank until the bearing is free from the cylinder block.



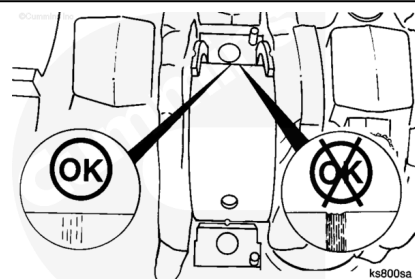
Clean and Inspect for Reuse

⚠ CAUTION ⚠

To reduce the possibility of damage to the crankshaft, do not use a scraper or a wire brush on the crankshaft journals.

Use a lint-free cloth to clean the crankshaft journal.

Inspect the crankshaft journal for damage.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

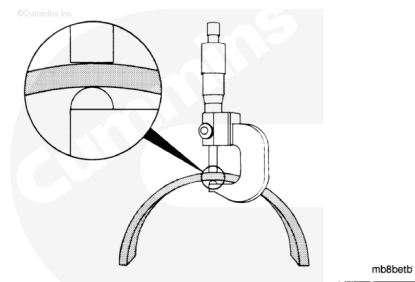
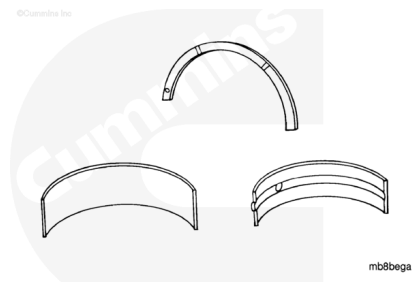
To reduce the possibility of bearing damage, do not use a scraper or wire brush on bearing surfaces.

Clean the bearings with solvent and dry with compressed air.

⚠ CAUTION ⚠

To reduce the possibility of damage to the bearing surfaces, use adequate professional tools and measuring techniques. Use of improper measuring tools or methods may cause local surface damages that can become the initiation point for bearing surface layer failure.

Use a ball-end micrometer to measure the bearing thickness at the wear location.



Main Bearing Thickness			
Main Bearings	mm		in
Standard	3.949	MIN	0.1555
	3.962	MAX	0.1560
Oversize 0.25 mm [0.010 in]	4.074	MIN	0.1604
	4.087	MAX	0.1609
Oversize 0.50 mm [0.020 in]	4.199	MIN	0.1653
	4.212	MAX	0.1658

Main Bearing Thickness			
Main Bearings	mm		in
Oversize 0.75 mm [0.030 in]	4.324	MIN	0.1702
	4.337	MAX	0.1707
Oversize 1.0 mm [0.040 in]	4.449	MIN	0.1752
	4.462	MAX	0.1757

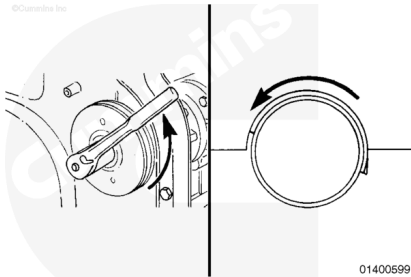
Install

⚠ CAUTION ⚠

Used bearings must be installed in their original location. Engine damage can result if bearings are not installed in the original position.

⚠ CAUTION ⚠

Do not use the bearing roll-out tool to install the upper bearings. The tool can slide under the bearing and engine damage can result.



Install the upper main bearing shells as far as possible by hand. The bearing will want to spring away from the crank. Use a flat piece of bar stock (approximately 3.175 x 38.1 mm [0.125 x 1.5 in]) to press the bearing against the crank while turning the crankshaft in the direction to rotate the bearing into place.

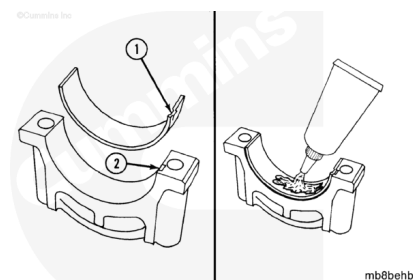
The bearing tang **must** fit into the slot in the bearing saddle to assure proper location of the bearing.

Use a lint-free cloth. Clean the lower main bearings, the lower thrust bearings, and the mounting surfaces.

Do **not** lubricate the back of the main bearings.

Align the tang (1) in the bearing with the slot (2) in the main bearing cap. Install the bearing. The end of the bearing **must** be even with the main bearing cap mounting surface.

Lubricate the bearing surface with clean engine oil.



mb8behb

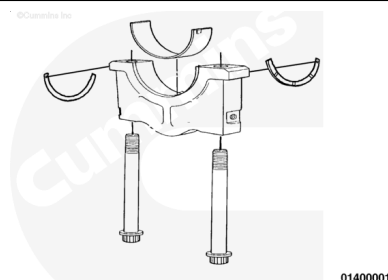
⚠ CAUTION ⚠

The grooves in the thrust bearings must point toward the crankshaft. The dowels that secure the bearings must not protrude above the bearing. Engine damage can result.

Lubricate the capscrew head and threads with clean 15W-40 engine oil. Allow the excess oil to drip off the capscrews before installing in the block.

Install the capscrews in the cap.

Install the two thrust bearings on the number 6 main bearing cap. Use the following procedure for removal of the thrust bearing. Refer to Procedure 001-007 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-007.html>).

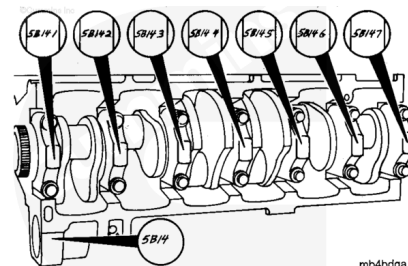


01400001

⚠ CAUTION ⚠

The numbers on the main bearing caps must be the same as the numbers on the block. If the main bearing caps are not installed in the same location, engine damage can result.

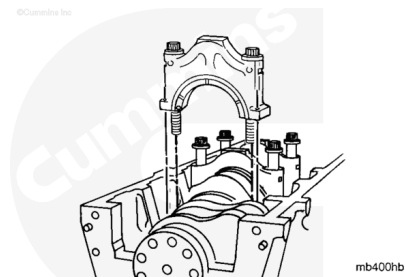
Match the number on the main bearing caps with the correct location in the block. The side of the cap marked "F" should face the front of the engine.



mb4bdga

⚠ CAUTION ⚠

Make sure the side of the cap and bearing with the bearing locating tang is toward the tang in the block. If the main bearing caps are not positioned correctly, engine damage can result.



mb400hb

⚠ CAUTION ⚠

Do not hit the main bearing caps with a hammer. The bearing shells can fall out.

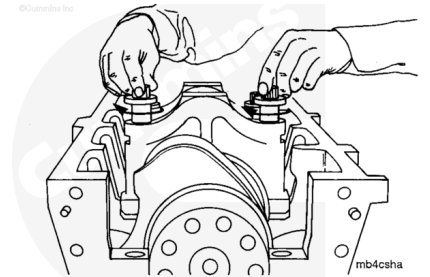
Lubricate the bearing surface with engine oil. The bearing shells **must** be firmly seated in the cap and the correct capscrew and washer combination in position. Install the main bearing caps.

⚠ CAUTION ⚠

Do not rotate the crankshaft until all of the main bearing caps are pulled to the block. Damage will result if the bearings move out of location.

Install the capscrews.

Turn each capscrew until it touches the main bearing cap.



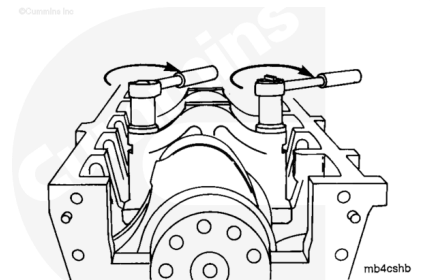
⚠ CAUTION ⚠

Do not use an impact wrench. The main bearing shells can fall out.

Use both of the capscrews to pull the main bearing cap into position.

Use two wrenches and tighten both capscrews at the same time.

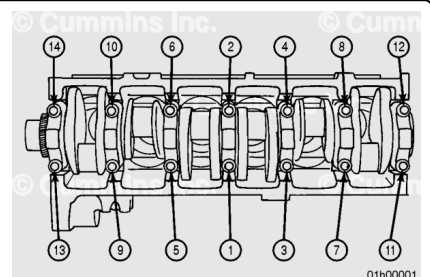
Check to be sure the number 7 cap is even with the back of the block and that the thrust bearings are still in place.



Use the following steps to tighten the capscrews in the sequence shown.

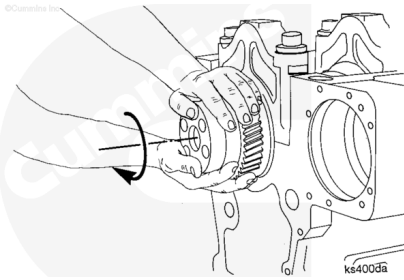
Torque Value:

1. 285 n•m [210 ft-lb]
2. 580 n•m [428 ft-lb]
3. Tighten 90°



Turn the crankshaft by hand. If the pistons have **not** been removed, use the barring mechanism.

The crankshaft will turn freely if the main bearings are installed correctly.



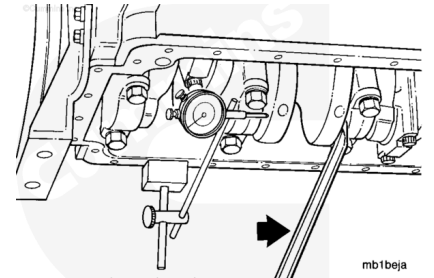
Use a dial indicator to measure the crankshaft end clearance.

Crankshaft End Clearance

New Minimum	New Maximum	Worn Limit
0.14 mm [0.0055 in]	0.32 mm [0.0125 in]	0.60 mm [0.0236 in]

If the clearance is **not** within specifications, check for foreign material. Oversize thrust bearings are available to adjust the end clearance.

The upper and lower thrust bearing on the same side of the main caps **must** be the same thickness.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ CAUTION ⚠

Do not pry or bend the suction tube to get the support bracket capscrow holes to align. A bent suction tube can cause improper oil flow and cause engine damage.

- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/89/89-001-089.html)
- Install the suction tube. Refer to Procedure 001-089 in Section 1 (/qs3/pubsys2/xml/en/procedures/89/89-001-

089.html).

- Install the oil pan and related components. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/89/89-007-025.html)
- Fill the engine with clean 15W-40 lubricating oil. Use the following procedure for engine oil capacity specifications. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/89/89-007-037.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to 70°C [160°F] coolant temperature and check for leaks.